

# Chapter 3

## Vectors

A vector is a quantity that has direction as well as magnitude. Quantities that only have magnitude are called scalars.

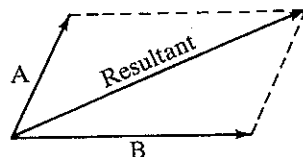
| Vectors                  | Scalars     |
|--------------------------|-------------|
| displacement             | distance    |
| velocity                 | speed       |
| acceleration             | time        |
| momentum                 | mass        |
| force                    | temperature |
| moment                   | charge      |
| flux density             | density     |
| electric field intensity | volume      |
|                          | area        |
|                          | energy      |
|                          | power       |

All vector quantities must involve a statement of direction. Vectors can be represented by an "arrow" in which the magnitude of the vector is given by its length and the direction is given by the direction of the arrow.

## Addition

The addition of vectors must take into account the directional factor and the result is best seen in a diagram.

All vectors are added by placing the tail of one to the tip of the first. The resultant is the vector which goes from the tail of the first vector added to the tip of the last added. The parallelogram rule is often used for adding vectors that act simultaneously. If force vector A is to be added to force vector B the resultant is as shown.



When vectors act at right angles to each other, the diagonal is easily found by Pythagoras' Theorem. The direction of the resultant is also easily found by trigonometry.

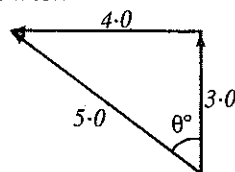
### Example 1

What is the sum of two displacements 3.0 m north and 4.0 m west?

$$\theta = \tan^{-1} \left( \frac{4}{3} \right)$$

$$= 53^\circ$$

The resultant displacement is 5.0 m at north 53° west.



## Subtraction

The difference between two vectors can be found by reversing the direction of the vector being subtracted and then adding it. This is because the negative

value of a vector is another vector with the same magnitude but having the opposite direction.

$$A - B = A + (-B)$$

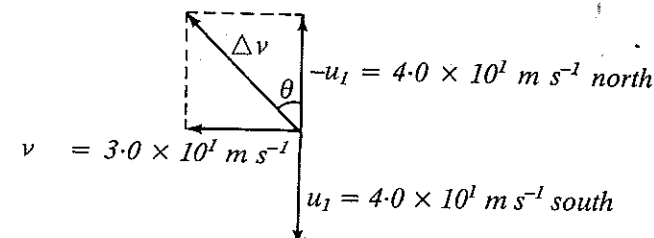
### Example 2

What is the change in velocity when a cricket ball, moving at  $4.0 \times 10^1 \text{ m s}^{-1}$  south, is hit over square leg which is due west at  $3.0 \times 10^1 \text{ m s}^{-1}$ ?

$$u = 4.0 \times 10^1 \text{ m s}^{-1} \text{ south} \quad \Delta v = v - u = v + (-u)$$

$$v = 3.0 \times 10^1 \text{ m s}^{-1} \text{ west} \quad \Delta v = 3.0 \times 10^1 \text{ m s}^{-1} \text{ west} + 4.0 \times 10^1 \text{ m s}^{-1} \text{ north}$$

$$\Delta v = ?$$



$$\Delta v = \sqrt{30^2 + 40^2} = 50$$

$$\theta = \tan^{-1} \left( \frac{30}{40} \right) = 37^\circ$$

The change in velocity is  $5.0 \times 10^1 \text{ m s}^{-1}$  at north 37° west.

## Set 3

- Add the following displacements.
  - 3 m north and 4 m south
  - 5.0 m north and 3.0 m west
- Add the forces:
  - 5.0 N west 5.0 N north
  - 6.0 N west + 6.0 N south + 6.0 N north-west
- Find the sum of the following velocities.
  - 12 m s<sup>-1</sup> north and 5.0 m s<sup>-1</sup> east
  - 6.0 m s<sup>-1</sup> north, 4.0 m s<sup>-1</sup> east, 3.0 m s<sup>-1</sup> south
- A boat can travel at 3.0 m s<sup>-1</sup> in still water and wishes to travel directly across a river with a current of 1.0 m s<sup>-1</sup>. What direction upstream does the boat need to be steered?
- If the boat in question 4 heads directly across the river which is 1.0 km wide how far downstream will it reach the opposite bank?
- A plane is heading due north at 120 km h<sup>-1</sup> in an easterly crosswind of 50.0 km h<sup>-1</sup>. What is the actual direction along the ground?
- Subtract the following vectors.
  - 4 m north from 3 m north
  - 4 m north from 3 m south
  - 4 m north from 3 m east
  - 3.0 m east from 5.0 south

- 8 If a car travelling south at  $40 \text{ m s}^{-1}$  slows down to  $20 \text{ m s}^{-1}$ , what is the change in velocity?
- 9 A batsman hits a cricket ball travelling towards him at  $30 \text{ m s}^{-1}$  over square leg at  $50 \text{ m s}^{-1}$ . If the cricketer is facing north and square leg is west of him, what is the change in velocity of the ball?
- 10 If a car travelling south at  $50 \text{ km h}^{-1}$  rounds a bend and travels east at the same speed, what is its change in velocity?

### Answers

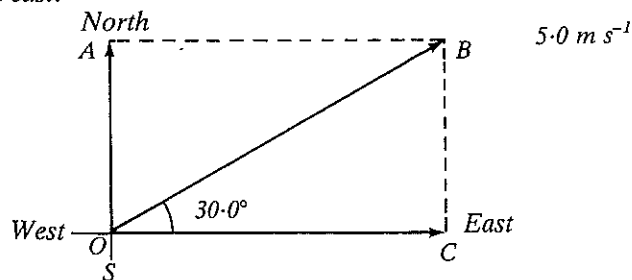
- 1 (a)  $1 \text{ m south}$  (b)  $5.8 \text{ m north } 31^\circ \text{ west}$
- 2 (a)  $7.1 \text{ N north-west}$  (b)  $1.0 \times 10^1 \text{ N west } (1.0 \times 10^1)^\circ \text{ south}$
- 3 (a)  $13 \text{ m s}^{-1} \text{ north } 23^\circ \text{ east}$  (b)  $5 \text{ m s}^{-1} \text{ north } 53^\circ \text{ east}$
- 4  $71^\circ \text{ upstream to current}$  5  $330 \text{ m}$  6  $\text{north } 23^\circ \text{ west}$
- 7 (a)  $1 \text{ m south}$  (b)  $7 \text{ m south}$   
(c)  $5 \text{ m south } 40^\circ \text{ east}$  (d)  $5.6 \text{ m south } 31^\circ \text{ west}$
- 8  $20 \text{ m s}^{-1} \text{ north}$  9  $60 \text{ m s}^{-1} \text{ north } 60^\circ \text{ west}$
- 10  $20 \text{ m s}^{-1} \text{ north-east}$

### Resolution of vectors

Just as two vectors can be added, a single vector can be split into two parts or components. It is often easier to deal with the components of several vectors in two mutually perpendicular directions, since the components in one direction have no effect on the component in the other direction.

#### Example 3

What are the components of a velocity of  $5.0 \text{ m s}^{-1}$  north  $60.0^\circ$  east in the directions north and east?



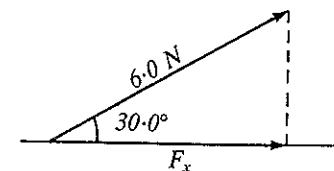
The two components must add up to the origin vector. Thus completing the rectangle  $OACB$  the components are north,  $OA$ , and east,  $OC$ .

Using trigonometry  $OB = 5.0 \text{ m s}^{-1}$   $\angle BOC = 30.0^\circ$   
 north component  $= 5.0 \sin 30.0^\circ = 2.5 \text{ m s}^{-1}$   
 east component  $= 5.0 \cos 30.0^\circ = 4.3 \text{ m s}^{-1}$

The north component is  $2.5 \text{ m s}^{-1}$  and the east component is  $4.3 \text{ m s}^{-1}$ .

#### Example 4

What is the horizontal component of a force of  $6.0 \text{ N}$  acting at  $30.0^\circ$  to the horizontal?



horizontal component is  $F_x$

$$\frac{F_x}{6.0} = \cos 30.0^\circ$$

$$\therefore F_x = 6.0 \cos 30.0^\circ$$

$$= 5.2 \text{ N}$$

The horizontal component is  $5.2 \text{ N}$ .

#### Set 4

- 1 What is the horizontal component of a force of  $11 \text{ N}$  acting at  $60.0^\circ$  to the horizontal?
- 2 What is the northerly component of a wind blowing at  $15 \text{ m s}^{-1}$  from the south-east?
- 3 What is the vector whose components are  $3 \text{ N}$  horizontally and  $4 \text{ N}$  vertically?
- 4 What are the vertical and horizontal components of a force of  $25 \text{ N}$  acting at  $30.0^\circ$  to the horizontal?
- 5 A plane taking off leaves the runway at  $32^\circ$  to the horizontal, travelling at  $180 \text{ km h}^{-1}$ . How long will it take to climb to an altitude of  $1000 \text{ m}$ ?
- 6 What is the acceleration of a small object down a smooth plane inclined at  $20.0^\circ$  to the horizontal? Take the acceleration due to gravity  $g = 9.80 \text{ m s}^{-2}$ .
- 7 (a) Add the vertical components of the forces acting on the point shown in the diagram.  
(b) Which way will it move?
- 8 What is the total force acting in question 7?
- 9 A boat crossing a river which flows at  $5.0 \text{ m s}^{-1}$  south heads directly across at  $2.0 \text{ m s}^{-1}$ . If the river is  $1.0 \text{ km}$  wide:  
(a) how long does the boat take to cross, and  
(b) how far downstream does it reach the opposite bank?

